

SIF Verification — PFDavg

SIF TagSIF-001
Target SIL2

Parameter	Sensor (S)	Logic Solver (LS)	Final Element (FE)	Unit
Architecture (1oo1/1oo2/2oo2/3oo3)	2oo2	1oo1	1oo1	—
λDU (Dangerous Undetected)	5,70E+01	1,80E+01	2,60E+01	FIT
λDD (Dangerous Detected)	1,70E+01	1,70E+01	5,70E+01	FIT
TI (Proof Test Interval)	8 760	8 760	8 760	h
MTTR (Mean Time To Restoration)	8	8	24	h
MT (Mission Time)	87 600	87 600	87 600	h
PTC (Proof Test Coverage)	90,0%	90,0%	90,0%	fraction
β (Common Cause Undetected)	10,0%	5,0%	10,0%	fraction
βD (Common Cause Detected)	5,0%	2,5%	5,0%	fraction

Note: λDU and λDD entered in FIT (failures per 10⁹ h). Internal calculation converts to 1/h.

Internal calculations

λDU (1/h)	5,70E-08	1,80E-08	2,60E-08
λDD (1/h)	1,70E-08	1,70E-08	5,70E-08
PFD 1oo1	4,74E-04	1,50E-04	2,18E-04
PFD 1oo2	2,51E-05	3,96E-06	1,15E-05
PFD 2oo2	5,00E-04	1,58E-04	2,30E-04
PFD 2oo3	2,54E-05	3,99E-06	1,15E-05

Selected PFDavg per subsystem	5,00E-04	1,50E-04	2,18E-04
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Contribution to total (%)	57,6%	17,3%	25,1%
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SIF RESULT

Total PFDavg (SIF)	8,67E-04
Achieved SIL	3
Risk Reduction Factor	1 153
Target met?	YES